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Practical - 6 - Hamming code

AIM: Write a program to implement error detection and correction using HAMMING code concept. Make a test run to input data stream and verify error correction feature.

Error correction in Data Link Layer.

Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is transmitted from the sender to the receiver.

Create sender program with below feature.

1) Input to sender file should be a text of any length. Program convert text to binary.

2) Apply hamming code concept on the binary data and add redundant bit to it.

3) Save the output in a file called channel.

Create a receiver program with below feature.

1) Read encoded data from the channel file.

2) Check for errors using Hamming code.

3) Display the error position if found.

4) Remove the redundant bits and display the original text

Student observation:

Code:

1) Convert the input list to binary.
2) Add redundant bits using Hamming code.
3) Store the encoded data in the channel file
4) Check for errors at the receiver.
5) Correct single_bit errors and display the original list.

Input:

HELLO

Output:

* Encoded data stored in the channel file.
* Error detected and corrected.
* Original message displayed: HELLO.

Result:

Thus, Hamming code - error and detection was ~~done~~ successfully done.